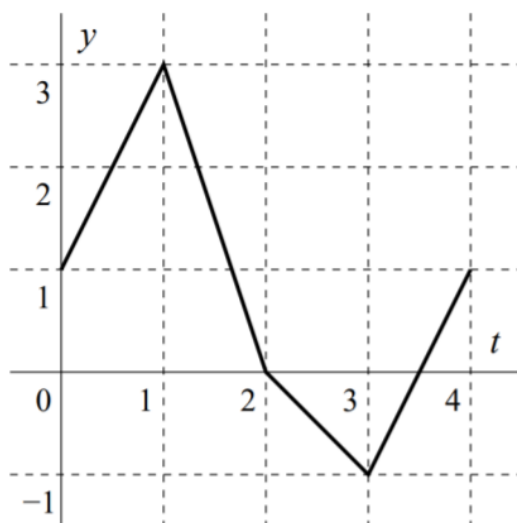


Workshop 13

Problem 1

Keywords: Fundamental Theorem of Calculus Part I, Graphs, Interpretation of Derivative and Integral.

The graph of f is shown below. The function $F(x)$ is defined by $F(x) = \int_0^x f(t)dt$ for $0 \leq x \leq 4$.



- Find $F(0)$ and $F(4)$.
- Find $F'(1)$ and $F'(3)$.
- Find the intervals of increase, decrease of $F(x)$, and its maximal value on the interval $[0, 4]$.

Problem 2

Keywords: Fundamental Theorem of Calculus Parts I and II

The goal of this problem is to calculate $\frac{d}{dx} \int_{2x}^{x^2} \ln(t)dt$ using the Fundamental Theorem of Calculus, Part I (FTCI) and then to double-check this calculation by using the Fundamental Theorem of Calculus, Part II (FTCII). Note that this second method is rarely possible: usually an FTCII-type question will involve an integral that you do not know how to calculate.

- Use FTCI to compute $\frac{d}{dx} \int_1^{x^2} \ln(t)dt$ and $\frac{d}{dx} \int_1^{2x} \ln(t)dt$.
- Use part (a) to compute $\frac{d}{dx} \int_{2x}^{x^2} \ln(t)dt$.

- (c) Compute $\frac{d}{dt}t \ln(t)$ and use your result to determine $\int \ln(t)dt$.
- (d) Compute $\int_{2x}^{x^2} \ln(t)dt$ using FTCII.
- (e) Take the derivative of your answer in part (d) to verify your answer from part (b).

Problem 3

Keywords: Substitution

- (a) For the indefinite integral $\int \frac{3x^2 dx}{4 + 100x^6}$, perform the substitution $u = x^6$. Is this substitution helpful to compute the indefinite integral?
- (b) Compute the indefinite integral $\int \frac{3x^2 dx}{4 + 100x^6}$ by performing the substitution $u = 5x^3$.
- (c) Suppose you first try to compute the above indefinite integral by performing the substitution $u = x^3$. You would have to perform a second substitution in order to evaluate the integral. What is this second substitution?
- (d) Performing a certain substitution, the integral $\int \frac{2x dx}{\sqrt[3]{x^2 + 1}}$ can be written $\int 3u du$. Find which substitution is used to do this.

Extra Problem (*for practice, no need to submit*)

Let f be a continuous function, and let F be an antiderivative of f . Suppose that $\int_0^1 f(x)dx = 8$, $\int_0^3 f(x)dx = 5$, $F(3) = 4$ and $F(2) = -5$. Find $F(1)$ and $\int_0^2 f(x)dx$.